Problem · Unit 5 · 10 points · 60 min · Standard exam level

P04 — Radial tube on a turntable with increasing rate: Euler force, three regimes and the motor torque

A horizontal turntable turns about the fixed vertical axis $O\mathbf{k}$ of the inertial frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}\}$ ($\mathbf{k}$ vertical upward, gravity $-g\,\mathbf{k}$). A motor imposes the angle $\theta(t)$ between $\mathbf{i}_0$ and a radial line of the table, with

$$\dot{\theta} = \Omega(t) = \Omega_0 + \beta t, \qquad \theta(0) = 0$$

*($\Omega_0 > 0$ and β constants). A smooth, thin, straight tube is fixed on the table along a diameter of the table, from the rim at $x = -L$ through O to the rim at $x = L$. A bead P of mass m moves inside the tube and is joined to O by a spring of stiffness k and **zero natural length**. The frame $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}\}$ turns with the table, $\mathbf{i}_1$ along the tube and $\mathbf{j}_1 = \mathbf{k} \times \mathbf{i}_1$; x is the signed coordinate of P along $\mathbf{i}_1$ (so $x < 0$ means on the other side of O). The table and the tube have moment of inertia I_T about the axis.*

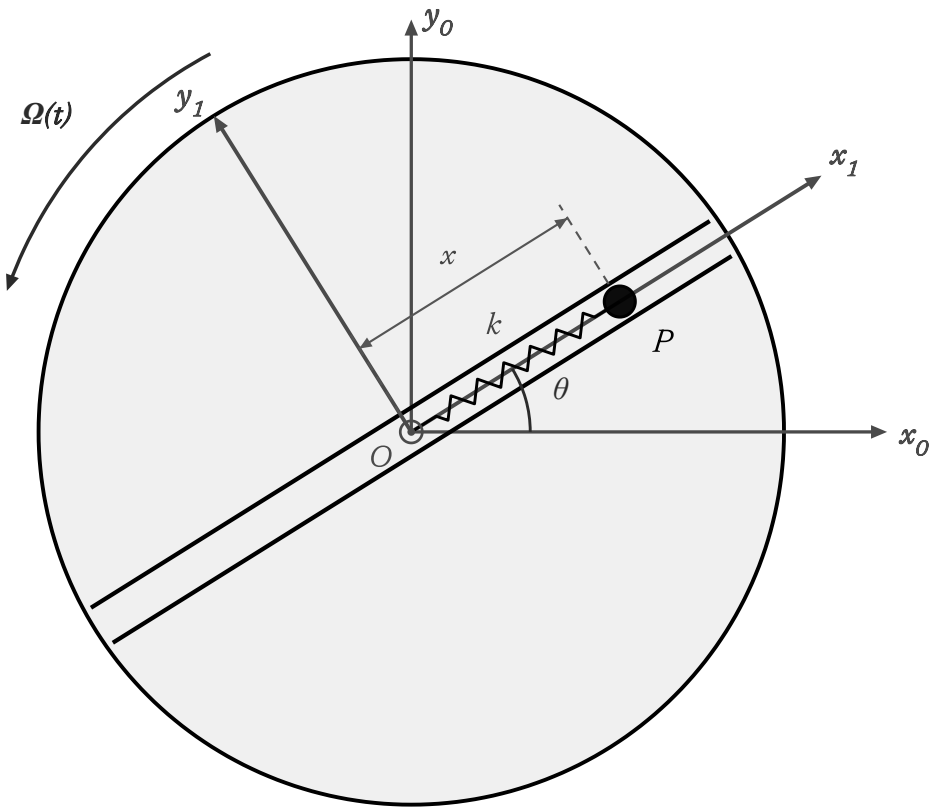


Figure 1. Top view: turntable with rate $\Omega(t) = \dot{\theta} = \Omega_0 + \beta t$ about the vertical axis $O\mathbf{k}$ (out of the page); smooth tube along $\mathbf{i}_1$, bead P at distance x from O, spring of stiffness k and zero natural length.

(The scalar k is the stiffness; the bold $\mathbf{k}$ is the vertical unit vector.)

Setup

- Data: $m, k, g, \Omega_0, \beta, I_T, L$.
- Numerical parts (e), (f): $m = 0.2$ kg, $k = 20$ N/m (so $\omega_n = \sqrt{k/m} = 10$ rad/s), $\beta = 0$, $x(0) = x_0 = 0.1$ m, $\dot{x}(0) = v_0 = 0$, $L = 0.5$ m, $g = 9.81$ m s^{-2}, and $\Omega_0 = 6, 10$ or 12.5 rad/s.
- Hypotheses: point particle, no friction in the tube, S_0 inertial, the motor delivers whatever torque is needed to keep $\Omega(t)$.

Part (a) · 1 point

Degrees of freedom and frames

Count the degrees of freedom of P. Give $[{}_0R_1]$, $\boldsymbol{\omega}_{10}$ and $\boldsymbol{\alpha}_{10}$.

Step 1. Count

In S_1 the tube is the line $y_1 = 0$, $z_1 = 0$: two holonomic constraints, so $3 - 2 = 1$ degree of freedom, x (Notes §2.7). In S_0 the tube turns with the prescribed $\theta(t)$, so the constraints depend explicitly on time (rheonomic).

Step 2. Frames

$\mathbf{i}_1 = \cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0, \mathbf{j}_1 = -\sin\theta\,\mathbf{i}_0 + \cos\theta\,\mathbf{j}_0$: the columns of $[{}_0R_1]$ above, a simple rotation about $\mathbf{k}$ (Notes §1.5) with $\det = \cos^2\theta + \sin^2\theta = 1$ and $RR^T = I$. Then $\boldsymbol{\omega}_{10} = \dot{\theta}\,\mathbf{k} = \Omega(t)\,\mathbf{k}$ and $\boldsymbol{\alpha}_{10} = \dot{\Omega}\,\mathbf{k} = \beta\,\mathbf{k}$.

KEY RESULTS

- 1 degree of freedom, x; constraints $y_1 = 0, z_1 = 0$ (rheonomic in S_0).
- $[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\boldsymbol{\omega}_{10} = \Omega(t)\,\mathbf{k}$, $\boldsymbol{\alpha}_{10} = \beta\,\mathbf{k}$.

Part (b) · 1.5 points

Kinematics

Write the relative velocity and acceleration of P and its absolute acceleration in $\mathcal{B}_1$, (i) with the Coriolis theorem and (ii) by differentiating $\mathbf{r} = x\,\mathbf{i}_1$ directly in S_0.

Step 1. Coriolis theorem

O is fixed, so (Notes §3.2) $\mathbf{a} = \mathbf{a}_{rel} + \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + 2\boldsymbol{\omega} \times \mathbf{v}_{rel}$, with $\mathbf{r} = x\,\mathbf{i}_1$ and $\mathbf{k} \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k} \times \mathbf{j}_1 = -\mathbf{i}_1$:

$$\boldsymbol{\alpha}\times\mathbf{r}=\beta x\,\mathbf{j}_1,\qquad \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r})=\Omega\,\mathbf{k}\times\Omega x\,\mathbf{j}_1=-\Omega^2 x\,\mathbf{i}_1,\qquad 2\boldsymbol{\omega}\times\mathbf{v}_{rel}=2\Omega\dot{x}\,\mathbf{j}_1$$

$$\mathbf{a}=(\ddot{x}-\Omega^2 x)\,\mathbf{i}_1+(2\Omega\dot{x}+\beta x)\,\mathbf{j}_1$$

Step 2. Direct differentiation

$\mathbf{v}=\dfrac{d}{dt}(x\,\mathbf{i}_1)=\dot{x}\,\mathbf{i}_1+x\Omega\,\mathbf{j}_1$ (product rule). Again, with $\dfrac{d}{dt}(x\Omega)=\dot{x}\Omega+x\dot{\Omega}=\dot{x}\Omega+x\beta$:

$$\mathbf{a}=\ddot{x}\,\mathbf{i}_1+\dot{x}\Omega\,\mathbf{j}_1+(\dot{x}\Omega+x\beta)\,\mathbf{j}_1+x\Omega(-\Omega\,\mathbf{i}_1)$$

which is the same result. The term $\beta x\,\mathbf{j}_1$ is the tangential acceleration of the point of the table under P.

KEY RESULTS

- $\mathbf{v}_{rel}=\dot{x}\,\mathbf{i}_1$, $\mathbf{a}_{rel}=\ddot{x}\,\mathbf{i}_1$.
- $\mathbf{a}=(\ddot{x}-\Omega^2 x)\,\mathbf{i}_1+(2\Omega\dot{x}+\beta x)\,\mathbf{j}_1$.

Part (c) · 1.5 points

Forces and inertia forces

Write all the forces on P in S_1, including the three inertia forces.

Step 1. Applied forces and reaction

The spring has zero natural length, so its force is $-k\,\overrightarrow{OP}=-kx\,\mathbf{i}_1$. The weight is $-mg\,\mathbf{k}$. The tube is smooth: its reaction has no $\mathbf{i}_1$ component, $\mathbf{N}=N_y\,\mathbf{j}_1+N_z\,\mathbf{k}$.

Step 2. Inertia forces of S_1

From the terms of (b) multiplied by $-m$ (Notes §4.9):

$$\mathbf{F}_E=-m\beta x\,\mathbf{j}_1,\qquad \mathbf{F}_{cf}=m\Omega^2 x\,\mathbf{i}_1,\qquad \mathbf{F}_{cor}=-2m\Omega\dot{x}\,\mathbf{j}_1$$

The Euler (azimuthal) force exists only because the rotation accelerates; it points against the sense of rotation for $\beta>0$.

KEY RESULTS

- $\mathbf{F}_k = -kx\,\mathbf{i}_1$, $\mathbf{W} = -mg\,\mathbf{k}$, $\mathbf{N} = N_y\,\mathbf{j}_1 + N_z\,\mathbf{k}$.
- $\mathbf{F}_E = -m\beta x\,\mathbf{j}_1$ (Euler), $\mathbf{F}_{cf} = m\Omega^2 x\,\mathbf{i}_1$, $\mathbf{F}_{cor} = -2m\Omega\dot{x}\,\mathbf{j}_1$.

Part (d) · 1.5 points

Equation of motion and reactions

Obtain the equation of motion free of reactions, and N_y, N_z. Why does the equation contain Ω^2 but not β?

Step 1. Projections

$$\mathbf{i}_1:\ m\ddot{x} = -kx + m\Omega^2 x \qquad \mathbf{j}_1:\ 0 = N_y - m\beta x - 2m\Omega\dot{x} \qquad \mathbf{k}:\ 0 = N_z - mg$$

Step 2. Result

$$\ddot{x} + \left(\frac{k}{m} - \Omega(t)^2\right)x = 0, \qquad N_y = m(\beta x + 2\Omega\dot{x}), \qquad N_z = mg$$

Step 3. Why no β

The Euler force $-m\beta x\,\mathbf{j}_1$ and the Coriolis force are perpendicular to the tube: the wall absorbs them and they appear only in N_y. The angular acceleration still acts on x, but only through the time-dependent coefficient $\Omega(t)^2 = (\Omega_0 + \beta t)^2$. (In S_0 the same equation is the $\mathbf{i}_1$ component of $m\mathbf{a}$ of (b).)

KEY RESULTS

- $\ddot{x} + \left(\frac{k}{m} - \Omega(t)^2\right)x = 0$.
- $N_y = m(\beta x + 2\Omega\dot{x})$, $N_z = mg$.

Part (e) · 2 points

Constant rate: the three regimes

Take $\beta = 0$. Solve the equation of motion with $x(0) = x_0$, $\dot{x}(0) = v_0$ for $\Omega_0^2 < k/m$, $\Omega_0^2 = k/m$ and $\Omega_0^2 > k/m$, write the Jacobi integral, and evaluate each case for the data of the Setup ($\Omega_0 = 6$, 10, 12.5 rad/s).

Step 1. $\Omega_0^2 < k/m$

$\ddot{x} + \omega^2 x = 0$ with $\omega^2 = k/m - \Omega_0^2 > 0$ (Notes Ch. 5): $x = A\cos\omega t + B\sin\omega t$. $x(0) = A = x_0$; $\dot{x} = -A\omega\sin\omega t + B\omega\cos\omega t$ gives $\dot{x}(0) = B\omega = v_0$. Data, $\Omega_0 = 6$: $\omega = \sqrt{100 - 36} = 8$ rad/s, $x = 0.1\cos 8t$ m, period $2\pi/8 = 0.7854$ s. The spring wins over the centrifugal force.

Step 2. $\Omega_0^2 = k/m$ **(the trap)**

$\ddot{x} = 0$: integrating twice with Barrow's rule, $\dot{x} = v_0$ and $x = x_0 + v_0 t$. With $v_0 = 0$ the bead stays where it is, wherever that is: the spring force kx is exactly the centrifugal force $m\Omega_0^2 x$ at every x. With $v_0 \neq 0$ the bead drifts at constant relative speed. It is neither an oscillation of zero frequency nor an exponential escape. Data, $\Omega_0 = 10$: $x \equiv 0.1$ m.

Step 3. $\Omega_0^2 > k/m$

$\ddot{x} - \lambda^2 x = 0$ with $\lambda^2 = \Omega_0^2 - k/m$: $x = A\cosh\lambda t + B\sinh\lambda t$, with $x(0) = A = x_0$ and $\dot{x}(0) = B\lambda = v_0$. Writing it as $\tfrac{1}{2}(x_0 + v_0/\lambda)e^{\lambda t} + \tfrac{1}{2}(x_0 - v_0/\lambda)e^{-\lambda t}$, the growing exponential disappears only for $v_0 = -\lambda x_0$, when $x = x_0 e^{-\lambda t}$. Data, $\Omega_0 = 12.5$: $\lambda = \sqrt{156.25 - 100} = 7.5$ s^{-1}, $x = 0.1\cosh 7.5t$ m.

Step 4. Jacobi integral

Multiplying the equation by $\dot{x}$: $\dfrac{d}{dt}\left[\tfrac{1}{2}\dot{x}^2 + \tfrac{1}{2}\left(\dfrac{k}{m} - \Omega_0^2\right)x^2\right] = 0$, so

$$\tfrac{1}{2}\dot{x}^2 + \tfrac{1}{2}\left(\frac{k}{m} - \Omega_0^2\right)x^2 = \text{const}$$

the energy in S_1 with the spring and centrifugal potentials. $\mathbf{N}$, $\mathbf{F}_{cor}$ and $\mathbf{F}_E$ do no work in S_1 because they are orthogonal to $\mathbf{v}_{rel}$.

KEY RESULTS

- $\Omega_0^2 < \omega_n^2$: $x = x_0\cos\omega t + \dfrac{v_0}{\omega}\sin\omega t$, $\omega = \sqrt{\omega_n^2 - \Omega_0^2}$.
- $\Omega_0^2 = \omega_n^2$: $x = x_0 + v_0 t$ (every position is a neutral relative equilibrium).
- $\Omega_0^2 > \omega_n^2$: $x = x_0\cosh\lambda t + \dfrac{v_0}{\lambda}\sinh\lambda t$, $\lambda = \sqrt{\Omega_0^2 - \omega_n^2}$; it decays only if $v_0 = -\lambda x_0$.
- Data: $\omega = 8$ rad/s, $x = 0.1\cos 8t$ m, period 0.7854 s; $x \equiv 0.1$ m; $\lambda = 7.5$ s^{-1}, $x = 0.1\cosh 7.5t$ m.

Part (f) · 1 point

Reaching the end of the tube

For $\Omega_0 = 12.5$ rad/s, $\beta = 0$ and the data of the Setup, find the time at which the bead reaches the rim ($x = L$), its relative speed then, and the reaction there.

Step 1. Time

$0.1 \cosh 7.5 t_L = 0.5 \Rightarrow \cosh 7.5 t_L = 5 \Rightarrow t_L = \dfrac{\operatorname{arccosh} 5}{7.5} = \dfrac{2.2924}{7.5} = 0.3057$ s.

Step 2. Speed

$\dot x = x_0 \lambda \sinh \lambda t$ with $\sinh \lambda t_L = \sqrt{25 - 1} = \sqrt{24}$: $\dot x = 0.1 \cdot 7.5 \cdot \sqrt{24} = 3.674$ m/s. Check with the Jacobi integral: $\dot x^2 = \lambda^2 (L^2 - x_0^2) = 56.25 \cdot 0.24 = 13.5$ m^2 s^{-2}.

Step 3. Reaction

$N_y = m(0 + 2\Omega_0 \dot x) = 2 \cdot 0.2 \cdot 12.5 \cdot 3.674 = 18.37$ N, about nine times the weight $N_z = mg = 1.962$ N.

KEY RESULTS

- $t_L = \dfrac{\operatorname{arccosh} 5}{7.5} = 0.3057$ s; $\dot x = 0.75\sqrt{24} = 3.674$ m/s.
- $N_y = 2m\Omega_0 \dot x = 18.37$ N, $N_z = mg = 1.962$ N.

Part (g) · 1.5 points

Motor torque and power balance

With the angular-momentum theorem about the axis, find the torque M the motor must deliver to keep $\Omega(t)$, and check it against N_y. Write the power balance of the bead in S_0. For the case of (f), give M (beyond $I_T \beta$) and the power at $x = L$, and the work of the motor on the bead between x_0 and L.

Step 1. Angular momentum of the bead about the axis

$\mathbf{r} \times m\mathbf{v} = x\,\mathbf{i}_1 \times m(\dot x\,\mathbf{i}_1 + \Omega x\,\mathbf{j}_1) = m x^2 \Omega\,\mathbf{k}$, so $H_z = m x^2 \Omega$. By the product rule:

$$\frac{dH_z}{dt} = m\big(2x\dot x\,\Omega + x^2 \dot\Omega\big) = m\big(2x\dot x\,\Omega + x^2 \beta\big)$$

Step 2. Torque on the bead

Moments about the axis (Notes §4.8): the spring force passes through O; $\mathbf{r}\times\mathbf{W} = x\,\mathbf{i}_1\times(-mg\,\mathbf{k}) = mgx\,\mathbf{j}_1$ and $\mathbf{r}\times N_z\mathbf{k}$ are horizontal; only $\mathbf{r}\times N_y\mathbf{j}_1 = xN_y\,\mathbf{k}$ counts. Hence $\dfrac{dH_z}{dt} = xN_y$, and with N_y of (d): $xN_y = mx(\beta x + 2\Omega\dot{x})$, the same expression as Step 1. The two routes agree.

Step 3. Motor torque

The table receives the motor torque M and the moment $-xN_y$ of the bead's push on the tube: $I_T\beta = M - xN_y$, so

$$M = I_T\beta + mx\big(\beta x + 2\Omega\dot{x}\big)$$

Even at constant rate ($\beta = 0$) the motor must supply $2m\Omega x\dot{x}$ whenever the bead moves radially.

Step 4. Power balance in S_0

$T = \tfrac{1}{2}m(\dot{x}^2 + \Omega^2x^2)$, $V = \tfrac{1}{2}kx^2$. Differentiating, $\dfrac{d}{dt}(T+V) = m\dot{x}\ddot{x} + m\Omega\beta x^2 + m\Omega^2x\dot{x} + kx\dot{x}$, and with $m\ddot{x} = -kx + m\Omega^2x$ from (d): $\dfrac{d}{dt}(T+V) = 2m\Omega^2x\dot{x} + m\Omega\beta x^2 = \Omega\,xN_y$. This is the power of $N_y\mathbf{j}_1$ on the absolute velocity, $N_y\,\Omega x$ (Notes §4.4): in S_0 the reaction of the moving tube works; in S_1 it does not.

Step 5. Numbers at $x = L$ (case (f), $\beta = 0$)

$xN_y = 0.5\cdot 18.37 = 9.186$ N m and the power is $\Omega_0 xN_y = 12.5\cdot 9.186 = 114.8$ W. Since $\beta = 0$, the work of the motor on the bead is $\displaystyle\int M\Omega_0\,dt = \Omega_0\int\frac{dH_z}{dt}dt = \Omega_0\,[H_z]_0^{t_L} = \Omega_0\big(H_z(t_L) - H_z(0)\big) = m\Omega_0^2(L^2 - x_0^2) = 0.2\cdot 156.25\cdot 0.24 = 7.5$ J. Check: T goes from 0.15625 to $\tfrac{1}{2}\cdot 0.2\,(13.5 + 39.0625) = 5.25625$ J and V from 0.1 to 2.5 J, so $\Delta(T+V) = 5.1 + 2.4 = 7.5$ J.

KEY RESULTS

- $H_z = mx^2\Omega$, $\dfrac{dH_z}{dt} = m(2x\dot{x}\Omega + x^2\beta) = xN_y$; $M = I_T\beta + mx(\beta x + 2\Omega\dot{x})$.
- $\dfrac{d}{dt}(T+V) = \Omega\,xN_y = M_{bead}\,\Omega$ (the reaction works in S_0, not in S_1).
- Case (f): $xN_y = 9.186$ N m, power 114.8 W, work $= \Omega_0\Delta H_z = m\Omega_0^2(L^2 - x_0^2) = 7.5$ J $= \Delta(T+V)$.

Interpretation and checks

- **Dimensional check:** k/m and Ω^2 are s^{-2}; $N_y = m(\beta x + 2\Omega\dot{x})$ is kg (s^{-2} m) = N; $H_z = mx^2\Omega$ is kg m^2 s^{-1}; M is N m and $M\Omega$ is W.
- **Limits:** $k = 0$ gives the classic bead on a rotating rod, $x = x_0\cosh\Omega_0 t$; $\Omega_0 = 0$, $\beta = 0$ gives the plain oscillator $\omega = \omega_n$. With $\beta > 0$ the coefficient $k/m - \Omega(t)^2$ eventually becomes negative: a bead that oscillates at first is pushed outwards once $\Omega(t) > \omega_n$.

- **Common mistakes in the exam:** (1) forgetting the Euler force $-m\boldsymbol{\alpha}\times\mathbf{r}$ (it changes N_y and the torque, not the equation of motion); (2) at $\Omega_0^2=k/m$ writing an exponential or a harmonic solution instead of $x=x_0+v_0t$; (3) claiming that the energy $T+V$ of the bead is conserved because the tube is smooth: in S_0 the tube moves and its reaction works; (4) computing the motor torque without the $I_T\beta$ term of the table, or with the wrong sign of the bead's push.